

Advanced Population Modeling : Reference Points

Gavin Fay

2025-11-13

Objectives

- ▶ Project updates
- ▶ Introduction to Reference Points
- ▶ Population Projections

(Biological) Reference Points (BRP)

- ▶ Generically, a metric of stock status from a biological perspective
- ▶ Often reflects combination of components of stock dynamics (growth, recruitment, mortality)
- ▶ 3 most common models for BRPs:
 1. Spawner-Recruit
 2. Production models
 3. Dynamic Pool (per-recruit)

Gabriel & Mace (1994)

Table 1. Summary of principal models that underlie biological reference points, and associated specification of age-structured and stock-recruitment data.

Age structure in population	S-R data required	S-R function required	Model type	Example citation	Reference points	Comments
Unknown	No	No	Surplus production	Schaefer, 1957 Prager, 1993	F_{MSY} , B_{MSY}	Very risk-prone without auxiliary data on recent relative recruitment
No (semelparous)	Yes	Yes	Spawner-recruit	Ricker, 1975	P_t , u_t	
Yes (iteroparous)	No	No	Dynamic pool, Y/R	Thompson and Bell, 1934	F_{max} , $F_{0.1}$	No information about reproductive dynamics
	By analogy	By analogy	Dynamic pool, SSB/R		$F_{20\%/SPR}$, $F_{35\%/SPR}$	No stock-recruitment relationship, except by analogy
	Yes	No	Dynamic pool, SSB/R	Shepherd, 1982	F_{med}	
	Yes	Yes	Dynamic pool, SSB/R	Mace, 1994	F_ϵ	
	Yes	Yes	Production	Sissenwine and Shepherd, 1987	F_{MSY} , B_{MSY}	

TLDR

Defining Biological Reference Points in a Dynamic Northeast U.S. Marine Environment

CINAR Workshop

January 8-9, 2024

Kerr, L.A., Jesse, J., Cadrin, S.X., editors



Production models

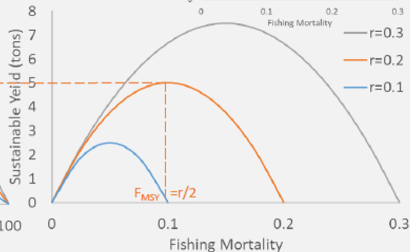
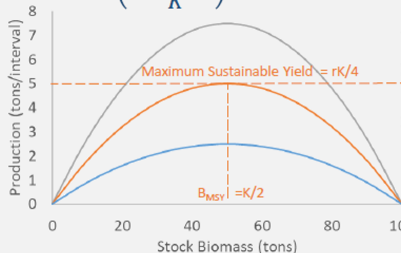
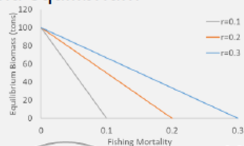
Fishery Application of Logistic Growth

- Graham (1935) – Density-dependent sustainable yield and equilibrium*

$$\frac{dB}{dt} = rB_t - \frac{r}{K}B_t^2 - Y_t \quad B^* = K - \left(\frac{K}{r}\right)F$$

$$\left(r - \frac{r}{K}B^*\right)B^* = FB^*$$

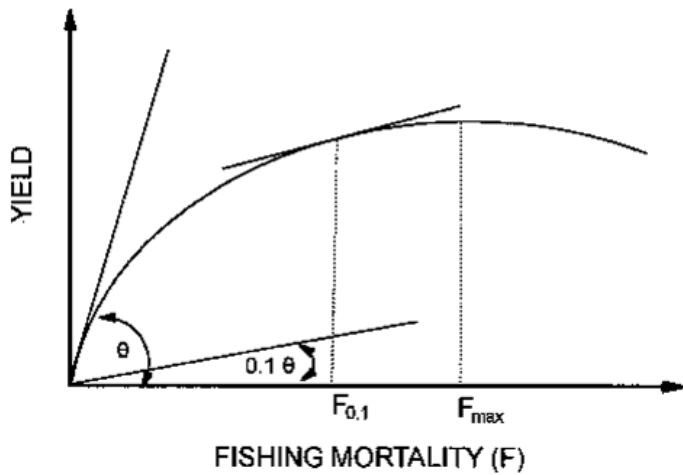
$$Y^* = FB^*$$



Reference points from Yield-per-recruit (YPR)

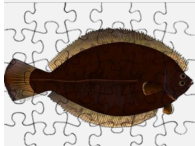
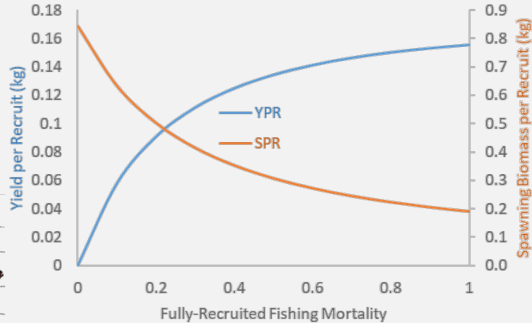
- ▶ F_{max} , fully-recruited F that produces maximum YPR.
- ▶ $F_{0.1}$, F corresponding to 10% slope of YPR curve at origin.

These BRPs are in context of growth overfishing.



Yield and Spawner per Recruit Calculations

- With little selectivity on immature ages, YPR is typically asymptotic, and $F_{\max}$ is poorly defined.
- $F_{40\%}=0.3$

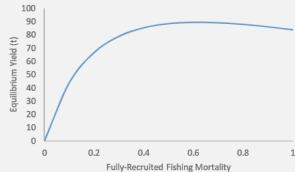
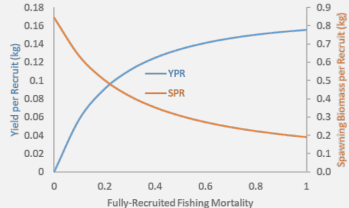


Spawning stock biomass per recruit models

- ▶ Spawning potential
- ▶ Under no fishing, 100% Spawning potential obtained.
- ▶ Quantify spawning stock biomass per recruit relative to unfished (Spawning Potential Ratio)
- ▶ e.g. F_{35} , the F that results in a SPR of 35% the unfished level.

Management Reference Points

- Reference points are based on long-term expectations (usually at equilibrium)
 - Yield-per-recruit ($F_{\max}$, $F_{0.1}$) – expected yield from an arbitrary number of recruits, fished at a constant F and selectivity pattern over its lifetime.
 - Spawner-per-recruit ($F_{\%MSP}$) – expected spawning potential (mature biomass, egg production, ...) from an arbitrary number of recruits, fished at a constant F and selectivity pattern over its lifetime.
 - Maximum Sustainable Yield (F_{MSY} , SSB_{MSY}) – expected yield (and stock) from fishing at a constant F (and selectivity) over generations

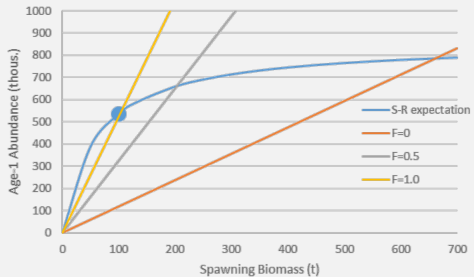
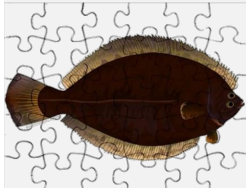


Age-structured FMSY

- ▶ Combine SPR with YPR
- ▶ Expected number of recruits from $SBPR(F)$ & knowledge of R/SSB
- ▶ Multiply by $YPR(F)$ to get the yield
- ▶ Growth and recruitment overfishing

'Self-Regenerating Model'

- Equilibrium expectations (e.g., $S_0=662$ t) can be derived graphically, by plotting stock-recruit expectations with SPR expectations, $R=S/(1/SPR_F)$.
- Note that assuming $h=0.75$ (132 k recruits at 588 t SSB) largely determines equilibrium expectations.

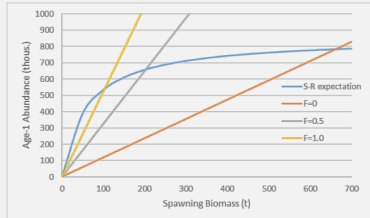
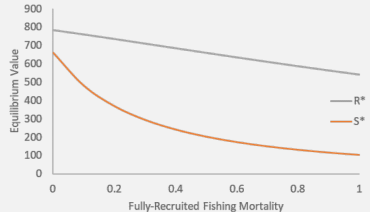


'Self-Regenerating Model'

- The same equilibrium expectations can be derived analytically by replacing S_t with $R^* SPR_F$.

$$R_t = \frac{4hR_0S_t}{S_0(1-h) + S_t(5h-1)} e^{\varepsilon_t}$$

$$R^* = \frac{4hR_0SPR_F - S_0(1-h)}{SPR_F(5h-1)} e^{\varepsilon_t}$$



In practice FMSY is hard to estimate

- ▶ principally rely on knowing say h
- ▶ More often use proxies for FMSY (e.g. F_{40})

Non-equilibrium

- ▶ These approaches are all equilibrium approaches
- ▶ Can be analytical, numerical, or projection-based
- ▶ Do not reflect process error variance
- ▶ Stochastic estimators for FMSY etc. also common
- ▶ Rely on projections, given assumptions about future dynamics
- ▶ e.g. Future recruitment, weight at age, etc.
- ▶ Take some quantile of the distribution over some time-frame

Lab exercise: Floundah per-recruit analysis

► Write functions in R that calculate:

1. Yield-Per-Recruit

$$YPR(F) = \sum_a w_a s_a F N_a (1 - \exp(-(s_a F + M))) / (s_a F + M)$$

2. Spawning Biomass Per Recruit

$$SPBR(F) = \sum_a m_a w_a N_a$$

$$N_a = N_{a-1} \exp(-(s_{a-1} F + M))$$

Lab exercise: Floundah per-recruit analysis

- Use the biology (& selectivity) data for floundah to find the values for:

1. F_{max}
2. $SBPR_0$
3. $SBPR$ as a function of F
4. $F_{40\%}$, i.e. F where $SBPR(F)/SBPR_0 = 0.4$